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Compute Project

# OCP MHS Modular Plug-and-Play Hardware Component Schema Documentation

Part of the Modular Hardware Systems (MHS) Rev 1.0 Family

Version 0.7.3

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# Version 0.7.3 Developer Guide

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**L** = Local, **R** = Required

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## Top-level Properties

**Specification:** This resource shall represent the hardware properties of an HPM or of a component assembly such as a backplane or riser that are necessary to enable plug-and-play management of that component by a DC-SCM module attached to the HPM or to an HPM bearing the component.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| _comment | string | | | This property shall contain a general comment about a data file |
| Copyright | string | | | This property shall contain a copyright statement for the FRU data. The omission of this property shall not imply that this FRU data is not copyrighted. |
| Version | string | | | This property shall contain the current file version for this FRU data. |
| SchemaVersion | string | | | This property shall contain the revision of the HardwareComponent schema that this FRU data was built against. |
| GitHubChecksum | string | | | This property shall contain the GitHub checksum for this FRU data. |
| Author | string | | | This property shall contain the author of this FRU data. |
| Component | Top-level Properties.Component | | | This resource shall contain information about a hardware component. |

### Top-level Properties.Component

**Specification:** This resource shall contain information about a hardware component.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| ComponentClass | string | Y | Y | This property shall contain the type of hardware that this component represents. Possible values: `HPM` (A host processor module, such as a DNO, FLW, or SDNO board), `Riser` (A riser), `Backplane` (A backplane), `PowerDistributionBoard` (A power distribution board), `PCIeSwitch`, `Other` (Any other component) |
| ComponentSubclass | string | Y | Y | This property shall contain the subtype of hardware that this component represents. |
| VendorID | string | Y | Y | This property shall contain the PCIe vendor ID assigned to the manufacturer for this peripheral. |
| UniqueID | string | Y | Y | This property shall contain a unique ID assigned by the manufacturer for this peripheral. Note that this is not necessarily a PCIe DeviceID. |
| PowerPolicy | string | | | This property shall contain the minimum system power state in which the component can be powered for full operation. Possible values: `G3`, `Pre-STBY`, `STBY`, `S5`, `S0` |
| MaximumPowerWatts | integer | | | This property shall contain the maximum wattage that the component can draw when fully powered. |
| PhysicalDimensions | Top-level Properties.Component.PhysicalDimensions | | | This property shall contain physical dimensions of the PCB for the component. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Connectors | Connectors | | | This property shall contain the hardware connectors that are present. |
| Devices | Devices | | | This property shall contain information about active and passive hardware devices. |
| Buses | Buses | | | This property shall contain information about the logical and physical buses. |

**Top-level Properties.Component.PhysicalDimensions**

**Specification:** This property shall contain physical dimensions of the PCB for the component.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| PCBWidthMM | integer | | | This property shall contain the width of the PCB in millimeters. |
| PCBLengthMM | integer | | | This property shall contain the length of the PCB in millimeters. |

## Connectors

**Specification:** This property shall contain the hardware connectors that are present.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| SOCs | array of Connectors_SOCs | | | This property shall contain information about SOC connectors. |
| MemorySubsystems | array of Connectors_MemorySubsystem | | | This property shall contain information about memory subsystem connectors. |
| Composites | array of Connectors_Composite | | | This property shall contain information about composite connectors. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| CoolingSubsystems | array of Connectors_CoolingSubsystem | | | This property shall contain information about cooling subsystem connectors. |
| Drives | array of Connectors_Drive | | | This property shall contain information about drive connectors. |
| Mxios | array of Connectors_Mxio | | | This property shall contain information about Mxio connectors. |
| PICPWRs | array of Connectors_PICPWR | | | This property shall contain information about PICPWR connectors. |
| OCPMezzanineSlots | array of Connectors_OCPMezzanineSlot | | | This property shall contain information about OCP mezzanine expansion slot connectors. |
| PowerSupplies | array of Connectors_PowerSupply | | | This property shall contain information about power supply connectors. |
| ControlPanels | array of Connectors_ControlPanel | | | This property shall contain information about control panel connectors. |
| PCIe_CEMs | array of Connectors_PCIe_CEM | | | This property shall contain information about PCIe CEM slot connectors. |
| PowerDistributionBoardManagements | array of Connectors_PowerDistributionBoardManagement | | | This property shall contain information about power distribution board management connectors. |
| RealTimeClockBatteries | array of Connectors_RealTimeClockBattery | | | This property shall contain information about real-time clock battery connectors. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Fans | array of Connectors_Fan | | | This property shall contain information about fan connectors. |
| SCIs | array of Connectors_SCI | Y | Y | This property shall contain information about SCI connectors. |
| IntrusionDetection | array of Connectors_IntrusionDetection | | | This property shall contain information about intrusion detection module connectors. |
| PhysicalUSBs | array of Connectors_PhysicalUSB | | | This property shall contain information about physical USB connectors. |
| DB9s | array of Connectors_DB9 | | | This property shall contain information about DB9 connectors. |
| OCPBoots | array of Connectors_OCPBoot | | | This property shall contain information about OCP boot connectors. |
| DisplayPorts | array of Connectors_DisplayPort | | | This property shall contain information about DisplayPort connectors. |
| OEM | array of Connectors_OEM | | | This property shall provide information for an OEM-specific connector. |

## Connectors_Composite

**Specification:** This property shall contain information about a composite connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |

| Property | Type | L | R | Specification |
| --- | --- | --- | --- | --- |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Type | string | | | This property shall contain the type of the composite. |
| DataFile | integer | | | This property shall contain the FRU index of the data file with JSON information specific to this composite. |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| PICPWRs | array of string | Y | Y | This property shall contain the names of the references to PICPWRs used by the composite. Each name shall correspond to an item in the `PICPWRs` section of the `Connectors` portion of this document. |
| MXIOs | array of string | Y | Y | This property shall contain the names of the references to MXIOs used by the composite. Each name shall correspond to an item in the `Mxios` section of the `Connectors` portion of this document. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | relative to an origin in the top left corner. |

## Connectors_ControlPanel

**Specification:** This property shall contain information about a control panel connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| Primary | boolean | | | This property shall indicate whether this is the primary control panel. |
| Sidebands | SidebandList | | | This property shall contain the informations about a set of sidebands. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| ConnectedBuses | ReferencedBusList | | | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |

## Connectors_CoolingSubsystem

**Specification:** This property shall contain information about a cooling subsystem connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| MaximumPowerWatts | integer | | | This property shall contain the maximum power for this cooling subsystem, in watts. |
| ConnectedBuses | ReferencedBusList | Y | Y | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |

## Connectors_DB9

**Specification:** This property shall contain information about a DB9 connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| ConnectedBuses | ReferencedBusList | | | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |

## Connectors_DisplayPort

**Specification:** This property shall contain information about a DisplayPort connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| ConnectedBuses | ReferencedBusList | | | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |

## Connectors_Drive

**Specification:** This property shall contain information about a drive connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| DriveType | string | | | This property shall contain the specific connector type for this drive. |
| HotPlugSupported | boolean | | | This property shall indicate whether this drive connector is wired to |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | support hot plug. |
| ConnectedBuses | ReferencedBusList | | | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |

## Connectors_Fan

**Specification:** This property shall contain information about a fan connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Pins | integer | | | This property shall contain the number of pins for the connector for this fan. |
| MaximumPowerWatts | integer | | | This property shall contain the maximum power for this fan, in watts. |
| Type | string | Y | Y | This property shall contain the type of connector for this fan. |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| HotPlugSupported | boolean | | | This property shall indicate whether this fan supports hot plugging. |
| ConnectedBuses | ReferencedBusList | Y | Y | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |

## Connectors_IntrusionDetection

**Specification:** This property shall contain information about an intrusion detection module connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| Type | string | Y | Y | This property shall contain the type of connector for this intrusion detection module. |
| StatusSignalNetname | string | | | This property shall contain the netname of the signal that provides the state of the intrusion detection sensor. |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |
| ConnectedBuses | ReferencedBusList | Y | Y | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |

## Connectors_MemorySubsystem

**Specification:** This property shall contain information about a memory subsystem connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| ChannelCount | integer | | | This property shall contain the number of channels supported by the memory subsystem. |
| SlotsPerChannel | unknown | | | This property shall contain the number of slots supported by each channel in the memory subsystem. If `SlotsPerChannel` consists of a single integer, that shall indicate that each channel supports the same number of slots. If `SlotsPerChannel` is an array, the array shall be of length `Channels` and each entry in the array shall be the number of slots supported by the corresponding channel. |
| MemoryTechnology | string | | | This property shall indicate the proximity of the memory subsystem |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | slot to its associated SOC. Possible values: `DDR4` (DDR4 SDRAM memory technology as defined in JEDEC specification JESD-4D.), `DDR5` (DDR5 SDRAM memory technology as defined in JEDEC specification JES-5C.01.), `DDR6` (DDR5 SDRAM memory technology as defined in a forthcoming JEDEC specification.), `Other` (Any other memory technology.) |
| Slots | array of Connectors_MemorySubsystem.Slots | Y | Y | This property shall contain information about an individual slot in the memory subsystem. |

**Connectors_MemorySubsystem.Slots**

**Specification:** This property shall contain information about an individual slot in the memory subsystem.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Proximity | string | | | This property shall indicate the proximity of the memory subsystem slot to its associated SOC. Possible values: `Near` (Close to the SOC.), `Far` (Further from the SOC, such that memory access performance is degraded.) |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| AssociatedSOC | string | | | This property shall contain an `Identifier` name for the SOC associated with this memory subsystem slot. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |
| ConnectedBuses | ReferencedBusList | Y | Y | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |

## Connectors_Mxio

**Specification:** This property shall contain information about an Mxio connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| SOCSocketAssociation | string | | | This property shall contain the name for the slot instance of the SOC Socket to which this MXIO is connected. |
| CommonClockingSupport | boolean | | | This property shall indicate whether the MXIO supports common clocking. |
| Type | string | Y | Y | This property shall contain the type of connector for this MXIO. |
| ConnectedBuses | ReferencedBusList | Y | Y | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| FlexIOs | array of FlexIO | Y | Y | This property shall contain the informations about the FLEXIOs used |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | by the MXIO. |
| Signals | array of Signal | | | This property shall contain information about the signals for this MXIO. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |

## Connectors_OCPBoot

**Specification:** This property shall contain information about an OCP boot connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |

| Property | Type | L | R | Specification |
| --- | --- | --- | --- | --- |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| Type | string | Y | Y | This property shall contain the type for the OCB boot connector. |
| Supports3_3V_AUX | boolean | | | No specification available |
| DualPortEnableSignalNetlistName | string | | | This property shall contain the netlist name for the dual port enable signal for the OCP boot connector. |
| LEDSignalNetlistName | string | | | This property shall contain the netlist name for the LED signal for the OCB boot connector. |
| PowerDisableSignalNetlistName | string | | | This property shall contain the netlist name for the power disable signal for the OCP boot connector. |
| SMResetNetlistName | string | | | This property shall contain the netlist name for the SM bus reset signal for the OCP boot connector. |
| ConnectedBuses | ReferencedBusList | Y | Y | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |
| FlexIOs | array of FlexIO | | | This property shall contain the informations about the FLEXIOs used |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | by the OCP Boot Connector. |

## Connectors_OCPMezzanineSlot

**Specification:** This property shall contain information about OCP mezzanine expansion slot connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Version | string | | | This property shall contain the version of the OCP NIC specification with which this OCP mezzanine expansion slot complies. |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| FormFactor | string | | | This property shall contain the form factor for this OCP mezzanine |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | expansion slot. Possible values: `SFF` (The Small Form Factor as defined in the OCP NIC 3.0 specification.), `TSFF` (The tall Small Form Factor as defined in the OCP NIC 3.0 specification.), `Other` (Any other form factor.) |
| ConnectedBuses | ReferencedBusList | Y | Y | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |
| PairedSlotForDoubleWide | string | | | This property shall contain an `Identifier` name for the OCP mezzanine expansion slot with which this OCP mezzanine expansion slot is paired for use as a double-wide connector in the DSFF or TDSFF form factor. |
| Signals | array of Signal | | | This property shall contain information about the signals for this OCP Mezzanine slot. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |

## Connectors_OEM

**Specification:** This property shall contain information about an OEM connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| Type | string | | | This property shall contain the type of this OEM connector. |
| PresenceDetectSignal | string | | | This property shall contain the netlist name of the signal (e.g., a GPIO) used to detect presence of a device connected via this non-standard connector. |
| PresenceDetectPolarity | string | | | This property shall contain the polarity indicating the active/present state of PresenceDetectSignal. Possible values: `active_high`, `active_low` |
| MaximumPowerWatts | integer | | | This property shall contain the maximum power draw for a device connected via this connector. |
| RequiredPowerState | RequiredPowerState | | | This property shall contain the minimum power state to utilize a bus. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| ConnectedBuses | ReferencedBusList | | | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |

## Connectors_PCIe_CEM

**Specification:** This property shall contain information about a PCIe CEM slot connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Type | string | Y | Y | This property shall contain the type for this CEM slot. |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| ConnectedBuses | ReferencedBusList | | | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |
| Signals | array of Signal | | | This property shall contain information about the signals for this PCIe CEM slot. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |

## Connectors_PICPWR

**Specification:** This property shall contain information about an PICPWR connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Type | string | | | This property shall contain the type of connector for this PICPWR. |
| AdjustedMaximumActualPowerSupportedWatts | integer | | | This property shall contain the adjusted maximum actual power supported for this PICPWR, in watts. |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| Sidebands | array of Connectors_PICPWR.Sidebands | Y | Y | This property shall contain the informations about the sidebands used by the PICPWR. |
| ConnectedBuses | ReferencedBusList | Y | Y | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |

**Connectors_PICPWR.Sidebands**

**Specification:** This property shall contain the informations about the sidebands used by the PICPWR.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this sideband. |

## Connectors_PhysicalUSB

**Specification:** This property shall contain information about a physical USB connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | within this document (excluding references to components). |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| Type | string | Y | Y | This property shall contain the USB type of the connector for this physical USB port. Possible values: `A` (USB type A connector as described in USB `A' Plug Form Factor)`, C `(USB type C connector as described in Universal Serial Bus Type-C Cable and Connector Specification)`, other` (Any other connector type) |
| Destination | string | | | This property shall contain the endpoint location of this physical USB port. Possible values: `Internal` (A USB port inside the system chassis.), `External` (A USB port external to the system chassis that can be accessed without opening the chassis.) |
| ConnectedBuses | ReferencedBusList | Y | Y | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |

## Connectors_PowerDistributionBoardManagement

**Specification:** This property shall contain information about a power distribution board management connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| ManagementType | string | | | This property shall contain the type for this power distribution management board connector. |
| Type | string | | | This property shall contain the connector type for this power distribution management board connector. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Direction | string | | | This property shall contain the form factor for this OCP mezzanine expansion slot. Possible values: `Upstream` (Up, toward the HPM), `Downstream` (Down, toward another PDB) |
| Sidebands | SidebandList | | | This property shall contain the informations about a set of sidebands. |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| ConnectedBuses | ReferencedBusList | Y | Y | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |

## Connectors_PowerSupply

**Specification:** This property shall contain information about a power supply connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Type | string | | | This property shall contain the type of connector for this power supply. |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| HotPlugSupported | boolean | | | This property shall indicate whether the power supply supports hot plugging. |
| ConnectedBuses | ReferencedBusList | Y | Y | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |

## Connectors_RealTimeClockBattery

**Specification:** This property shall contain information about a real-time clock battery connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | describe the component in the BOM list. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |
| ConnectedBuses | ReferencedBusList | | | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |

## Connectors_SCI

**Specification:** This property shall contain information about an SCI connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Revision | string | | | This property shall contain the revision of the Datacenter Secure Control Module Specification (DC-SCM) specification supported by this SCI. |

| Property | Type | L | R | Specification |
| --- | --- | --- | --- | --- |
| Version | string | | | This property shall contain the version of the Datacenter Secure Control Module Specification (DC-SCM) specification supported by this SCI. |
| CommonCircuitType | string | | | This property shall contain the number of nodes supported by this SCI. Possible values: `Single Node` (One SCI node), `Dual Node` (Two SCI nodes), `Other` (Any other number of SCI nodes) |
| BackupBatteryVoltageMeasurementCapable | boolean | | | This property shall indicate whether this fan supports measurement of the backup battery voltage. |
| ConnectedBuses | ReferencedBusList | | | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Signals | array of Signal | | | This property shall contain information about the signals for this SCI. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |

## Connectors_SOCs

**Specification:** This property shall contain information about an SOC connector.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| SupportedVendorName | string | | | This property shall contain the name of the vendor that this SOC connector supports. |
| SupportedVendorID | string | | | This property shall contain the PCIe VendorID for the vendor that this SOC connector supports. |
| Type | string | Y | Y | This property shall contain the type of the SOC. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| DimmChannelsSupported | integer | | | This property shall contain the number of dimm channels supported by the SOC. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| ConnectedBuses | ReferencedBusList | Y | Y | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |

## Devices

**Specification:** This property shall contain information about active and passive hardware devices.

**Array Item Properties:**

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Type | string | Y | Y | This property shall contain the type of the device. |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| Active | boolean | | | This property shall whether this is an active programmable device. |
| Primary | boolean | | | This property shall whether this is the primary active programmable device. For a passive device, this property shall be set to false or omitted. |
| DataFile | integer | | | This property shall contain the FRU index of the data file with JSON information specific to this device. |
| GPIOs | GPIOs | Y | | This property shall contain information about a set of GPIOs. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| CapacityBytes | number | | | This property shall contain the storage capacity for this device. |
| PhysicalSignals | array of Signal | Y | Y | This property shall contain information about the physical signals for this device. |
| ConnectedBuses | ReferencedBusList | Y | Y | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |
| OEM | Devices.OEM | | | This property shall provide additional OEM-specific information for a device. |

### Devices.OEM

**Specification:** This property shall provide additional OEM-specific information for a device.

No properties defined.

## Buses

**Specification:** This property shall contain information about the logical and physical buses.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| DisplayPort | array of Buses_DisplayPort | | | This property shall contain information about DisplayPort bus.s |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| ESPI | array of Buses_ESPI | Y | | This property shall contain information about ESPI bus.s |
| I2C | array of Buses_I2C | Y | Y | This property shall contain information about I2C bus.s |
| I3C | array of Buses_I3C | Y | | This property shall contain information about I3C bus.s |
| Intrusion | array of Buses_Intrusion | | | This property shall contain information about intrusion bus.s |
| JTAG | array of Buses_JTAG | Y | | This property shall contain information about JTAG bus.s |
| LTPI | array of Buses_LTPI | Y | | This property shall contain information about LTPI bus.s |
| MPESTI | array of Buses_MPESTI | Y | | This property shall contain information about MPESTI bus.s |
| NCSI_RBT | array of Buses_NCSI_RBT | Y | | This property shall contain information about NC-SI RBT bus.s |
| PCIe | array of Buses_PCIe | Y | Y | This property shall contain information about PCIe bus.s |
| PECI | array of Buses_PECI | | | This property shall contain information about PECI bus.s |
| QSPI | array of Buses_QSPI | | | This property shall contain information about QSPI bus.s |
| SGMII | array of Buses_SGMII | | | This property shall contain information about SGMII bus.s |
| SPI | array of Buses_SPI | Y | | This property shall contain information about SPI bus.s |
| SGPIO | array of Buses_SGPIO | Y | | This property shall contain information about SGPIO bus.s |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| UART | array of Buses_UART | | | This property shall contain information about UART bus.s |
| USB | array of Buses_USB | | | This property shall contain information about USB bus.s |
| OEM | array of Buses.OEM | | | This property shall provide information for an OEM-specific bus. |

## Buses.OEM

**Specification:** This property shall provide information for an OEM-specific bus.

No properties defined.

## Buses_DisplayPort

**Specification:** This property shall contain information about a DisplayPort bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this bus. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Segments | array of Buses_DisplayPort.Segments | | | This property shall contain information about DisplayPort segments. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

**Buses_DisplayPort.Segments**

**Specification:** This property shall contain information about DisplayPort segments.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| NetlistName | string | | | This property shall contain the name used in the netlist for this segment. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| MUXes | array of MUX | | | This object shall contain information about a MUX. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

## Buses_ESPI

**Specification:** This property shall contain information about an ESPI bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this bus. |
| MaxClockSpeedMHz | integer | | | This property shall contain the maximum clock speed for this ESPI bus. |
| AlertUsage | string | | | This property shall contain a description of how alerts are signaled on the bus. Possible values: `Embedded` (Alerts are encoded into data pins, |

| Property | Type | L | R | Specification |
| --- | --- | --- | --- | --- |
| | | | | signaling alerts via I/O[1].), `Discrete` (Alerts are encoded into a discrete signal on a dedicated alert pin.) |
| Density | string | | | This property shall contain the data transfer mode supported by this ESPI bus, as defined in the Enhanced Serial Peripheral Interface (eSPI) specification. Possible values: `Quad` (Four bi-directional I/O pins used for commands and data), `Dual` (Bi-directional I/O pins used for commands and data), `Single` (Unidirectional I/O pins used for commands and data) |
| Controller | string | | | This property shall contain the named component that receives interrupt requests and controls this bus. |
| Segments | array of Buses_ESPI.Segments | | | This property shall contain information about ESPI segments. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

**Buses_ESPI.Segments**

**Specification:** This property shall contain information about ESPI segments.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this segment. |
| MUXes | array of MUX | | | This object shall contain information about a MUX. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

## Buses_I2C

**Specification:** This property shall contain information about an I2C bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | Y | Y | This property shall contain the name used in the netlist for this bus. |
| VoltageTenthsVolts | integer | | | This property shall contain the voltage needed to drive this bus, in tenths of a volt. |
| RequiredPowerState | RequiredPowerState | Y | Y | This property shall contain the minimum power state to utilize a bus. |
| FrequencyKHz | integer | Y | Y | This property shall contain the clock frequency for this bus. |
| Description | string | | | This property shall contain a description of the intended usage for this bus. |
| Segments | array of Buses_I2C.Segments | Y | Y | This property shall contain information about I2C segments. |

**Buses_I2C.Segments**

**Specification:** This property shall contain information about I2C segments.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this segment. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |
| MUXes | array of MUX | | | This object shall contain information about a MUX. |
| Bridges | array of Buses_I2C.Segments.Bridges | | | This property shall contain information about I2C bridges in this I2C segment. |

#### Buses_I2C.Segments.Bridges

**Specification:** This property shall contain information about I2C bridges in this I2C segment.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| Version | string | | | This property shall contain the version for this bridge. |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| LinkedBuses | ReferencedBusList | | | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |

## Buses_I3C

**Specification:** This property shall contain information about an I3C bus.

| Property | Type | L | R | Specification |
| --- | --- | --- | --- | --- |
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this bus. |
| ProtocolType | string | | Y | This property shall contain the protocol type for this I3C bus. Possible values: `Basic` (Basic (royalty-free) I3C as defined in Specification for I3C Basic, Improved Inter Integrated Circuit -- Basic), `Standard` (I3C standard as defined in Specification for I3C Basic, Improved Inter Integrated Circuit -- Basic) |
| VoltageTenthsVolts | integer | | | This property shall contain the voltage needed to drive this I3C bus, in tenths of a volt. |
| RequiredPowerState | RequiredPowerState | | Y | This property shall contain the minimum power state to utilize a bus. |
| FrequencyKHz | integer | | Y | This property shall contain the frequency at which this I3C bus shall operate, in kilohertz. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Segments | array of Buses_I3C.Segments | Y | Y | This property shall contain information about I3C segments. |

**Buses_I3C.Segments**

**Specification:** This property shall contain information about I3C segments.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this segment. |
| Hubs | array of Buses_I3C.Segments.Hubs | Y | Y | This property shall contain information about I3C hubs. |
| MUXes | array of MUX | | | This object shall contain information about a MUX. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | Y | Y | This object shall contain a collection of references to connectors attached to a |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | bus as defined in the connectors section of this document. |

#### Buses_I3C.Segments.Hubs

**Specification:** This property shall contain information about I3C hubs.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Control | ChannelControl | | | This property shall contain information about mechanisms used to control this MUX or hub. |
| Ports | array of Buses_I3C.Segments.Hubs.Ports | Y | Y | This property shall contain information about I3C ports. |
| PropagationDelayNS | integer | | Y | This property shall contain the propagation delay for an I3C Hub, in nanoseconds. |
| Address | string | Y | Y | This property shall contain the I3C address at which this hub responds. |
| Models | Models | | | This property shall contain a list of the possible models for this component, |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| Description | string | | | This property shall contain a description of the intended usage for this hub. |

Buses_I3C.Segments.Hubs.Ports

**Specification:** This property shall contain information about I3C ports.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| ChannelConfig | ChannelConfig | | | This property shall contain information about a channel in a MUX or a hub. |
| VoltageTenthsVolts | integer | | | This property shall contain the voltage needed to drive this I3C port, in tenths of a volt. |
| TargetPortMode | string | | | This property shall contain the target mode for this I3C port. Possible values: `Basic` (Basic port mode, as defined in the I3C specification), `Standard` (MIPI port mode, as defined in the I3C specification) |
| Endpoint | Endpoint | | | This property shall contain a named Identifier for an endpoint attached to this component. |
| Description | string | | | This property shall contain a description of the intended usage for this port. |

## Buses_Intrusion

**Specification:** This property shall contain information about an intrusion bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this bus. |
| Segments | array of Buses_Intrusion.Segments | | | This property shall contain information about intrusion segments. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

**Buses_Intrusion.Segments**

**Specification:** This property shall contain information about intrusion segments.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this segment. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| MUXes | array of MUX | | | This object shall contain information about a MUX. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

## Buses_JTAG

**Specification:** This property shall contain information about a JTAG bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this bus. |
| Segments | array of Buses_JTAG.Segments | Y | Y | This property shall contain information about JTAG segments. A JTAG segment consisting of an identifier with no other properties defined shall be understood to be a terminator. |

**Buses_JTAG.Segments**

**Specification:** This property shall contain information about JTAG segments. A JTAG segment consisting of an identifier with no other properties defined shall be understood to be a terminator.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this segment. |
| MUXes | array of MUX | | | This object shall contain information about a MUX. |
| ConnectedDevices | ConnectedDeviceList | Y | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |
| Terminated | boolean | | | This property shall indicate whether this JTAG bus is terminated. |

## Buses_LTPI

**Specification:** This property shall contain information about an LTPI bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this bus. |
| RequiredPowerState | RequiredPowerState | | Y | This property shall contain the minimum power state to utilize a bus. |
| MaxClockSpeedMHz | integer | | | This property shall contain the maximum clock speed for this LTPI bus. |
| Segments | array of Buses_LTPI.Segments | | | This property shall contain information about LTPI bus segments. |
| ConnectedDevices | ConnectedDeviceList | Y | Y | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | Y | Y | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

**Buses_LTPI.Segments**

**Specification:** This property shall contain information about LTPI bus segments.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this segment. |
| MUXes | array of MUX | | | This object shall contain information about a MUX. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

## Buses_MPESTI

**Specification:** This property shall contain information about an MPESTI bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this bus. |
| Segments | array of Buses_MPESTI.Segments | | | This property shall contain information about MPESTI bus segments. |
| ConnectedDevices | ConnectedDeviceList | Y | Y | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | Y | Y | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

**Buses_MPESTI.Segments**

**Specification:** This property shall contain information about MPESTI bus segments.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this segment. |
| MUXes | array of MUX | | | This object shall contain information about a MUX. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

## Buses_NCSI_RBT

**Specification:** This property shall contain information about an NC-SI RBT bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| NetlistName | string | | | This property shall contain the name used in the netlist for this bus. |
| Segments | array of Buses_NCSI_RBT.Segments | | | This property shall contain information about segments for this NC-SI RBT bus. |

**Buses_NCSI_RBT.Segments**

**Specification:** This property shall contain information about segments for this NC-SI RBT bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this segment. |
| MUXes | array of MUX | | | This object shall contain information about a MUX. |
| Isolation | Buses_NCSI_RBT.Segments.Isolation | | | This property shall contain information about isolation management for this NC-SI RBT bus. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

**Buses_NCSI_RBT.Segments.Isolation**

**Specification:** This property shall contain information about isolation management for this NC-SI RBT bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Channels | array of Buses_NCSI_RBT.Segments.Isolation.Channels | | | This property shall contain information about channels for isolation management. |
| IsolationControl | array of Buses_NCSI_RBT.Segments.Isolation.IsolationControl | | | This property shall contain information about control for isolation management. |

**Buses_NCSI_RBT.Segments.Isolation.Channels**

**Specification:** This property shall contain information about channels for isolation management.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Endpoint | string | | | This property shall contain information about the endpoint to which this channel is connnected. |

#### Buses_NCSI_RBT.Segments.Isolation.IsolationControl

**Specification:** This property shall contain information about control for isolation management.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Signal | string | | | This property shall contain the signal that controls this isolation mechanism. |
| Polarity | string | | | This property shall contain the polarity of the signal that controls this isolation mechanism. Possible values: `active_high` (Set the signal to high to disable isolation and connect to the endpoint.), `active_low` (Set the signal to low to disable isolation and connect to the endpoint.) |

## Buses_PCIe

**Specification:** This property shall contain information about a PCIe bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this bus. |
| Lanes | integer | Y | Y | This property shall contain the number of PCIe lanes for this bus. |
| Version | number | Y | Y | This property shall contain the highest generation version of PCIe supported by this bus. |
| Root | string | Y | Y | This property shall contain the location of the PCIe root. Possible values: `Host` (The bus is rooted at the host SOC.), `BMC` (The bus is rooted at the BMC.), `Other` (The bus is not rooted at either the host SOC or at the BMC.) |
| Fabric | Buses_PCIe.Fabric | Y | Y | This property shall contain information about bus fabrics. |

**Buses_PCIe.Fabric**

**Specification:** This property shall contain information about bus fabrics.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this fabric. |
| HostSOCRootPort | string | | | This property shall contain the assigned port for the SOC PCIe root complex enumeration. |
| FirstLane | integer | | | This property shall contain the first PCIe lane used for this fabric. |
| Width | integer | | | This property shall contain the number of PCIe lanes used for this fabric. |
| Segments | array of Buses_PCIe.Fabric.Segments | | | This property shall contain information about segments of a PCIe bus. |

#### Buses_PCIe.Fabric.Segments

**Specification:** This property shall contain information about segments of a PCIe bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this segment. |
| Switch | Buses_PCIe.Fabric.Segments.Switch | | | This property shall contain information about the switch for a PCIe segment. |
| Bridges | array of Buses_PCIe.Fabric.Segments.Bridges | | | This property shall contain information about the bridges attached to a PCIe segment. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

Buses_PCIe.Fabric.Segments.Switch

**Specification:** This property shall contain information about the switch for a PCIe segment.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| VendorID | string | | | This property shall contain the PCIe vendor ID for this PCIe switch. |
| SubvendorID | string | | | This property shall contain the PCIe subvendor ID for this PCIe switch. |
| DeviceID | string | | | This property shall contain the PCIe device ID for this PCIe switch. |
| SubdeviceID | string | | | This property shall contain the PCIe subdevice ID for this PCIe switch. |
| RootPort | string | | | This property shall contain the root port associated with this PCIe switch. |
| LaneMapping | array of integer | | | This property shall contain a mapping of the PCIe lanes for this PCIe switch. |
| HotPlugSupported | boolean | | | This property shall indicate whether the PCIe switch is hot-pluggable. |
| DMASupported | boolean | | | This property shall indicate whether the PCIe switch supports direct memory access. |
| CxlSupported | boolean | | | This property shall indicate whether the PCIe switch supports Compute |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | Express Link (CXL). |
| CxlVersionMinimum | string | | | This property shall contain the minimum version of the CXL specification supported by this PCIe switch. |
| LinkSpeedSupportedMaximumGBPerSec | integer | | | This property shall contain the maximum link speed supported by this PCIe switch for a single lane, in gigabytes per second. If this property is omitted, the maximum transfer speed supported by the PCIe generation shall be assumed. |
| BifurcationCapable | boolean | | | This property shall indicate whether the PCIe switch supports bifurcation of its PCIe lanes. |
| BifurcationMinimum | integer | | | This property shall contain the minimum lane width supported by this PCIe switch for bifurcation. This property shall not be present for switches that do not support bifurcation. |
| LatencyNS | integer | | | This property shall contain the latency of this PCIe switch, in nanoseconds. |
| Ports | array of Buses_PCIe.Fabric.Segments.Switch.Ports | | | This property shall contain the The ports attached to this PCIe switch. |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | this component. |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |

# Buses_PCIe.Fabric.Segments.Switch.Ports

**Specification:** This property shall contain the The ports attached to this PCIe switch.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Width | integer | | | This property shall contain the number of PCIe lanes used for this port. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Endpoints | array of Endpoint | | | This property shall contain the endpoints attached to this port. |
| Type | string | | | This property shall contain the type of this port. Possible values: `Ingress` (This port is an ingress port.), `Egress` (This port is an egress port.) |

Buses_PCIe.Fabric.Segments.Bridges

**Specification:** This property shall contain information about the bridges attached to a PCIe segment.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| VendorID | string | | | This property shall contain the PCIe vendor ID for this bridge. |
| SubvendorID | string | | | This property shall contain the PCIe subvendor ID for this bridge. |
| DeviceID | string | | | This property shall contain the PCIe device ID for this bridge. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| SubdeviceID | string | | | This property shall contain the PCIe subdevice ID for this bridge. |
| RootPort | string | | | This property shall contain the root port associated with this PCIe bridge. |
| Width | string | | | This property shall contain the width, in PCIe lanes, for this PCIe bridge. |
| LaneMapping | array of integer | | | This property shall contain a mapping of the PCIe lanes for this PCIe bridge. |
| HotPlugSupported | boolean | | | This property shall indicate whether the PCIe bridge is hot-pluggable. |
| DMASupported | boolean | | | This property shall indicate whether the PCIe bridge supports direct memory access. |
| CxlSupported | boolean | | | This property shall indicate whether the PCIe bridge supports Compute Express Link (CXL). |
| CxlVersionMinimum | string | | | This property shall contain the minimum version of the CXL specification supported by this PCIe bridge. |
| LinkSpeedSupportedMaximumGBPerSec | integer | | | This property shall contain the maximum link speed supported by this PCIe bridge, in gigabytes per second. If this property is omitted, the maximum transfer speed supported by the PCIe generation shall be assumed. |
| BifurcationCapable | boolean | | | This property shall indicate whether the PCIe bridge supports bifurcation of its PCIe lanes. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| BifurcationMinimum | integer | | | This property shall contain the minimum lane width supported by this PCIe bridge for bifurcation. This property shall not be present for bridges that do not support bifurcation. |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |
| Endpoint | Endpoint | | | This property shall contain a named Identifier for an endpoint attached to this component. |

## Buses_PECI

**Specification:** This property shall contain information about a PECI bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this bus. |
| Segments | array of Buses_PECI.Segments | | | This property shall contain information about PECI bus segments. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

**Buses_PECI.Segments**

**Specification:** This property shall contain information about PECI bus segments.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this segment. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

## Buses_QSPI

**Specification:** This property shall contain information about a QSPI bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this bus. |
| MaxClockSpeedMHz | integer | | | This property shall contain the maximum clock speed for this QSPI bus. |
| Density | string | | | This property shall contain the data transfer mode supported by this ESPI bus, as defined in the Enhanced Serial Peripheral Interface (eSPI) specification. Possible values: `Quad` (Four bi-directional I/O pins used for commands and data), `Dual` (Bi-directional I/O pins used for commands and data), `Single` (Unidirectional I/O pins used for commands and data) |
| Controller | string | | | This property shall contain the named component that receives interrupt requests and controls this bus. |
| Segments | array of Buses_QSPI.Segments | | | This property shall contain information about QSPI segments. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

**Buses_QSPI.Segments**

**Specification:** This property shall contain information about QSPI segments.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this segment. |
| MUXes | array of MUX | | | This object shall contain information about a MUX. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

## Buses_SGMII

**Specification:** This property shall contain information about an SGMII bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this bus. |
| Segments | array of Buses_SGMII.Segments | | | This property shall contain information about SGMII bus segments. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

**Buses_SGMII.Segments**

**Specification:** This property shall contain information about SGMII bus segments.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this segment. |
| MUXes | array of MUX | | | This object shall contain information about a MUX. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

## Buses_SGPIO

**Specification:** This property shall contain information about an SGPIO bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this bus. |
| Segments | array of Buses_SGPIO.Segments | | | This property shall contain information about SGPIO bus segments. |
| ConnectedDevices | ConnectedDeviceList | Y | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | Y | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

**Buses_SGPIO.Segments**

**Specification:** This property shall contain information about SGPIO bus segments.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this segment. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| MUXes | array of MUX | | | This object shall contain information about a MUX. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

## Buses_SPI

**Specification:** This property shall contain information about a SPI bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this bus. |
| MaxClockSpeedMHz | integer | | | This property shall contain the maximum clock speed for this SPI bus. |
| AlertUsage | string | | | This property shall contain a description of how alerts are signaled |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | on the bus. Possible values: `Embedded` (Alerts are encoded into data pins, signaling alerts via I/O[1].), `Discrete` (Alerts are encoded into a discrete signal on a dedicated alert pin.) |
| Density | string | | | This property shall contain the data transfer mode supported by this ESPI bus, as defined in the Enhanced Serial Peripheral Interface (eSPI) specification. Possible values: `Quad` (Four bi-directional I/O pins used for commands and data), `Dual` (Bi-directional I/O pins used for commands and data), `Single` (Unidirectional I/O pins used for commands and data) |
| InterruptRequest | string | | | This property shall contain the named component to which the IRQ[0] pin is connected. This property shall not be present if the IRQ[0] pin is not connected. |
| Controller | string | | | This property shall contain the named component that receives interrupt requests and controls this bus. |
| Segments | array of Buses_SPI.Segments | | | This property shall contain information about SPI segments. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

**Buses_SPI.Segments**

**Specification:** This property shall contain information about SPI segments.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this segment. |
| MUXes | array of MUX | | | This object shall contain information about a MUX. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

## Buses_UART

**Specification:** This property shall contain information about a UART bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this bus. |
| VoltageTenthsVolts | integer | | Y | This property shall contain the voltage needed to drive this UART bus, in tenths of a volt. |
| RequiredPowerState | RequiredPowerState | | Y | This property shall contain the minimum power state to utilize a bus. |
| Segments | array of Buses_UART.Segments | | Y | This property shall contain information about UART segments. |

**Buses_UART.Segments**

**Specification:** This property shall contain information about UART segments.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this segment. |
| MUXes | array of MUX | | | This object shall contain information about a MUX. |
| ConnectedDevices | ConnectedDeviceList | | | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | | | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |

## Buses_USB

**Specification:** This property shall contain information about a USB bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this bus. |
| SupportedVersion | string | | Y | This property shall contain the maximum supported USB version for this USB bus. |
| HostController | string | | Y | This property shall contain the entity that controls the bus Possible values: `Host` (This bus is controlled by the host SOC.), `BMC` (This bus is controlled by the BMC.) |
| VoltageTenthsVolts | integer | | Y | This property shall contain the voltage needed to drive this USB bus, in tenths of a watt. |
| RequiredPowerState | RequiredPowerState | | Y | This property shall contain the minimum power state to utilize a bus. |
| Description | string | | | This property shall contain a description of the intended usage for this bus. |
| Segments | array of Buses_USB.Segments | | | This property shall contain information about segments for this USB bus. |

**Buses_USB.Segments**

**Specification:** This property shall contain information about segments for this USB bus.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this segment. |
| ConnectedDevices | ConnectedDeviceList | Y | Y | This property shall contain information about devices connected to a bus. |
| Connectors | AttachedConnectorList | Y | Y | This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document. |
| MUXes | array of MUX | | | This object shall contain information about a MUX. |
| Hubs | array of Buses_USB.Segments.Hubs | Y | Y | This property shall contain information about hubs for this USB bus segment. |
| Bridges | array of Buses_USB.Segments.Bridges | Y | Y | This property shall contain information about bridges for this USB bus segment. |

**Buses_USB.Segments.Hubs**

**Specification:** This property shall contain information about hubs for this USB bus segment.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Control | ChannelControl | | | This property shall contain information about mechanisms used to control this MUX or hub. |
| Ports | array of Buses_USB.Segments.Hubs.Ports | | | This property shall contain information about ports for this USB bus hub. |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Description | string | | | This property shall contain a description of the intended usage for this hub. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |

Buses_USB.Segments.Hubs.Ports

**Specification:** This property shall contain information about ports for this USB bus hub.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| ChannelConfig | ChannelConfig | | | This property shall contain information about a channel in a MUX or a hub. |
| VoltageTenthsVolts | integer | | | This property shall contain the voltage needed to drive this USB port, in tenths of a volt. |
| Endpoint | Endpoint | | | This property shall contain a named Identifier for an endpoint attached to this component. |

**Buses_USB.Segments.Bridges**

**Specification:** This property shall contain information about bridges for this USB bus segment.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| Version | string | | | This property shall contain the version of USB supported by this USB bridge. |
| LinkedBuses | ReferencedBusList | | | This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document. |

# Common Definitions

## AttachedConnectorList

**Specification:** This object shall contain a collection of references to connectors attached to a bus as defined in the connectors section of this document.

**Array Item Properties:**

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Type | string | Y | Y | This property shall contain the type of the bus being referenced. The type name shall match the top-level object name within the HPM `Connectors` section in this document. |
| ChipSelect | number | | | This property shall contain the chip selection pin needed to access this device. |
| Endpoint | string | Y | Y | This property shall contain the name of the instance of the connector being referenced. The instance name shall match the `Identifier` property name of the item within the array corresponding to the connector type. |
| MMBICapable | boolean | | | This property shall indicate whether the connector supports MMBI via the attached bus. |
| Width | integer | | | This property shall contain the number of lanes used for this connector. |

## ChannelConfig

**Specification:** This property shall contain information about a channel in a MUX or a hub.

**Array Item Properties:**

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| ControlPin | string | | | This property shall identify a pin needed to select this MUX or hub |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | channel. |
| Setting | number | | | This property shall identify the setting needed on a pin corresponding to `ControlPin` to select this MUX or hub channel. |

## ChannelControl

**Specification:** This property shall contain information about mechanisms used to control this MUX or hub.

**Array Item Properties:**

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| ControlPin | string | Y | Y | This property shall identify a pin used to control this MUX channel or hub. |
| Signal | ChannelControl.Signal | Y | Y | This resource shall identify the signal used to control the pin in `ControlPin`. |

### ChannelControl.Signal

**Specification:** This resource shall identify the signal used to control the pin in `ControlPin`.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Type | string | Y | Y | This resource shall identify the type locating signal the used to control the pin in `MUXPin`. |
| TypeID | string | Y | Y | This property shall identify the specific device housing the signal used to control the pin in `MUXPin`. |
| Subtype | string | Y | Y | This resource shall identify the subtype locating the signal used to control the pin in `MUXPin`. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| SubtypeID | string | Y | Y | This resource shall identify the specific signal used to control the pin in `MUXPin`. |

## ComponentAlias

**Specification:** This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).

**Type:** string

## ConnectedDeviceList

**Specification:** This property shall contain information about devices connected to a bus.

**Array Item Properties:**

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Endpoint | string | Y | Y | This property shall contain a named Identifier for this instance of a connected device. This named Identifier shall be unique across all identified items within this document (excluding references to items). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Address | string | Y | Y | This property shall contain the bus-specific hardware address needed to access a connected device. |

| Property | Type | L | R | Specification |
| --- | --- | --- | --- | --- |
| SecondaryAddress | string | | | This property shall contain the secondary bus-specific hardware address needed to access a connected device. |
| RandomAccessCapable | boolean | | | This property shall indicate whether the BMC can access memory on this device in random fashion through this connection. |
| ChipSelect | number | | | This property shall contain the chip selection pin needed to access this device. |
| HostAccessible | boolean | | | This property shall indicate whether the host can access this device. |

## Direction

**Specification:** This property shall contain the direction of a signal, virtual wire, or GPIO.

**Type:** string

**Allowed Values:** "input", "output", "bidir"

## Endpoint

**Specification:** This property shall contain a named Identifier for an endpoint attached to this component.

**Type:** string

## FlexIO

**Specification:** This property shall contain the informations about a FLEXIO.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this FLEXIO. |

## GPIOs

**Specification:** This property shall contain information about a set of GPIOs.

**Array Item Properties:**

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| StartBit | integer | | | This property shall contain the start bit for this GPIO. |
| WidthBits | integer | | | This property shall contain the width in bits for this GPIO. |
| Description | string | | | This property shall describe the intended use for this GPIO. |
| Direction | Direction | | | This property shall contain the direction of a signal, virtual wire, or GPIO. |

## MUX

**Specification:** This object shall contain information about a MUX.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Subtype | string | | | This property shall contain the type of this MUX. Possible values: `BUS_MUX` (A MUX that switches bus segments.), `SIGNAL_MUX` (A MUX that switches generic named signal pairs.), `BUFFER` (A buffer/repeater device.), |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | `TRANSCEIVER` (A transceiver device.), `LEVEL_TRANSLATOR` (A voltage level translator device.), `REDRIVER_CROSSPOINT_SWITCH` (A redriver/crosspoint switch device capable of routing multiple signal types (e.g. DisplayPort/USB3) to shared physical ports.) |
| SilkscreenName | SilkscreenName | | | This property shall contain the name silkscreened onto the HPM board for the component. |
| ReferenceDesignator | ReferenceDesignator | | | This property shall contain the reference designator used to logically describe the component in the BOM list. |
| Description | string | | | This property shall contain a description of this MUX. |
| Channels | array of MUX.Channels | Y | Y | This property shall contain information about channels in this MUX. |
| Control | ChannelControl | Y | | This property shall contain information about mechanisms used to control this MUX or hub. |
| InputSelector | ChannelControl | | | This property shall contain information about mechanisms used to control this MUX or hub. |
| Enable | ChannelControl | | | This property shall contain information about mechanisms used to control this MUX or hub. |
| Models | Models | | | This property shall contain a list of the possible models for this component, each as reported by the manufacturer. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Manufacturers | Manufacturers | | | This property shall contain a list of the names of possible manufacturers for this component. |
| PropagationDelayNS | integer | | | This property shall contain the propagation delay for this MUX, in nanoseconds. |
| PhysicalLocation | PhysicalLocation | | | This object shall contain the physical location of a component board, relative to an origin in the top left corner. |
| IsolationCapable | boolean | | | This property shall indicate whether the MUX can isolate voltage for channels. |
| LevelShiftingCapable | boolean | | | This property shall indicate whether the MUX is capable of shifting between voltage levels. |
| FaultDetectionCapable | boolean | | | This property shall indicate whether the MUX is capable of detecting faults. |
| FaultClearSignalCapable | boolean | | | This property shall indicate whether the MUX is capable of utilizing a fault clear signal. |
| RiseTimeNS | integer | | | This property shall contain the rise time for this MUX, in nanoseconds. |
| FallTimeNS | integer | | | This property shall contain the fall time for this MUX, in nanoseconds. |
| I2CI3CAddress | string | | | This property shall contain the I2C or I3C address for this MUX. This property shall be omitted if the MUX is not on an I2C or I3C bus. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| I2CI3CSecondaryAddress | string | | | This property shall contain the secondary I2C or I3C address for this MUX. This property shall be omitted if the MUX is not on an I2C or I3C bus. |

**MUX.Channels**

**Specification:** This property shall contain information about channels in this MUX.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| ChannelConfig | ChannelConfig | Y | Y | This property shall contain information about a channel in a MUX or a hub. |
| Endpoint | Endpoint | Y | Y | This property shall contain a named Identifier for an endpoint attached to this component. |
| VoltageTenthsVolts | integer | | | This property shall contain the voltage needed to drive this MUX channel, in tenths of a volt. |
| IOType | string | | | This property shall contain the type of I/O used for this MUX channel. Possible values: `push-pull` (Push-pull |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | dual-level output), `open drain` (Open drain I/O that drives to ground), `open collector` (Open collector I/O that drives high) |

## Manufacturers

**Specification:** This property shall contain a list of the names of possible manufacturers for this component.

Array of simple items.

## Models

**Specification:** This property shall contain a list of the possible models for this component, each as reported by the manufacturer.

Array of simple items.

## NamedComponent

**Specification:** This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).

**Type:** string

## PhysicalLocation

**Specification:** This object shall contain the physical location of a component board, relative to an origin in the top left corner.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| UpperXMils | integer | | | This object shall contain the upper x-axis corner of the bounding box surrounding the component, in millimeters. |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| UpperYMils | integer | | | This object shall contain the upper y-axis corner of the bounding box surrounding the component, in millimeters. |
| LowerXMils | integer | | | This object shall contain the lower x-axis corner of the bounding box surrounding the component, in millimeters. |
| LowerYMils | integer | | | This object shall contain the lower y-axis corner of the bounding box surrounding the component, in millimeters. |

## ReferenceDesignator

**Specification:** This property shall contain the reference designator used to logically describe the component in the BOM list.

**Type:** string

## ReferencedBusList

**Specification:** This object shall contain a collection of references to hardware or logical buses as defined in the buses section of this document.

**Array Item Properties:**

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Type | string | Y | Y | This property shall contain the type of the bus being referenced. The type name shall match the top-level object name within the top-level buses section in this document. |
| Identifier | string | Y | Y | This property shall contain the name of the instance of the bus type being |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | referenced. The instance name shall match the named `Identifier` or `Alias` property of the item within the array corresponding to the bus type. |
| SpecBusName | string | | | This property shall contain the name used in the spec to reference this bus. |
| Signals | array of Signal | | | This property shall contain information about the signals for this referenced bus. |

### RequiredPowerState

**Specification:** This property shall contain the minimum power state to utilize a bus.

**Type:** string

**Allowed Values:** "G3", "Pre-STBY", "STBY", "S5", "S0", "Other"

### SidebandList

**Specification:** This property shall contain the informations about a set of sidebands.

**Array Item Properties:**

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| Identifier | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | within this document (excluding references to components). |
| NetlistName | string | | | This property shall contain the name used in the netlist for this sideband. |

## Signal

**Specification:** This property shall contain information about a signal.

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| ConnectorName | string | | | This property shall contain the name of the connector that this signal uses. |
| Alias | ComponentAlias | | | This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Type | string | Y | Y | This property shall contain the type of the connector in which the physical signal is being referenced. The type name shall match the top-level object name within the `HPM` `Connectors` section in this document. |
| TypeID | NamedComponent | Y | Y | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| Subtype | string | | | This property shall contain the type of the connector in which the physical signal is being referenced. The type |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | name shall match the subtype's object name within the connector type corresponding to the `Type` property in the `HPM` `Connectors` section in this document. This property shall not be present unless `Type` is also present. |
| SubtypeID | NamedComponent | | | This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components). |
| SubtypeIDPrefix | string | | | This property shall contain the portion of the identifier that is common across a set of physical signals. This prefix should include everything up to the netlist name for the associated signals. When the `PhysicalPinPrefix` and pin numbers are appended to this property, the resulting identifier shall form the complete `SubtypeID` for a physical signal in the set. Each physical signal identifier in the set shall be unique across all named components within this document (excluding references to components). This property is mutually exclusive with the `SubtypeID` property and shall not be present if `SubtypeID` is present, or vice versa. |
| NetlistName | string | | | This property shall contain the Netlist name this physical signal reference. |
| NetlistNamePrefix | string | | | This property shall contain the portion of the Netlist name that is common |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | across a set of physical signals. When the `PhysicalPinPrefix` and pin numbers are appended to this property, the resulting names shall form the Netlist name for a physical signal in the set. This property is mutually exclusive with the `NetlistName` property and shall not be present if `NetlistName` is present, or vice versa. This property shall only be present if `SubtypeIDPrefix` is also present. |
| PhysicalPin | string | | | This property shall contain the name of the physical pin for this physical signal reference. |
| PhysicalPinPrefix | string | | | This property shall contain the portion of the physical pin name that is common across a set of physical signals. When pin numbers are appended to this property, the resulting names shall form the physical pin names for physical signals in the set. This property is mutually exclusive with the `PhysicalPin` property and shall not be present if `PhysicalPin` is present, or vice versa. This property shall only be present if `SubtypeIDPrefix` is also present. |
| FirstPhysicalPin | integer | | | This property shall contain the first physical pin number for a set of physical signal pins. This property is mutually exclusive with the `PhysicalPin` property and shall not be present if `PhysicalPin` is present, or vice versa. This property shall only |

| Property | Type | L | R | Specification |
|---|---|---|---|---|
| | | | | be present if `SubtypeIDPrefix` is also present. If this property is not present, but `PhysicalPinPrefix` is present, then the default value of 0 shall be used for `FirstPhysicalPin. |
| PhysicalPinCount | integer | | | This property shall contain the number of consecutive physical pin numbers in a set of physical signal pins. This property is mutually exclusive with the `PhysicalPin` property and shall not be present if `PhysicalPin` is present, or vice versa. This property shall only be present if `SubtypeIDPrefix` is also present |

## SilkscreenName

**Specification:** This property shall contain the name silkscreened onto the HPM board for the component.

**Type:** string